15 Amir has x nuts and y bolts in a box.

Amir takes 6 nuts and 6 bolts from the box.
The ratio of the number of nuts to the number of bolts now in the box is 1 : 2

Amir then takes 8 more nuts and 8 more bolts from the box.
The ratio of the number of nuts to the number of bolts now in the box is 2 : 5

Find, in its simplest form, the ratio x : y

After6 removed: $(x-6) : (y-6) = 1 : 2$

1 mark
Associate each removal with its own ratio.

$2(x-6) = y-6 \rightarrow 2x-y = 6$

After14 removed: $(x-14) : (y-14) = 2 : 5$

$5(x-14) = 2(y-14) \rightarrow 5x-2y = 42$

Double first equation: $4x-2y = 12$

1 mark
Cross-multiply both ratios.

Subtract: $x = 30$

1 mark
Eliminate y.

$2(30)-y = 6$, so $y = 54$

1 mark
Substitute to find y.

$x : y = 30 : 54 = 5 : 9$

1 mark
Simplify the ORIGINAL counts, not the remaining counts.

5 : 9

(Total for Question 15 is 5 marks)

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